



Gan.AI Enables Real-Time Creation of Personalized Videos to Drive Brand Engagement

Gan.AI reduces latency for custom video creation using multi-model endpoints on Amazon SageMaker, increasing customer stickiness and helping brands enrich marketing efforts.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

50 seconds

Average time to create custom video

3.000



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



1 million

unique videos created in 3 hours



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



Low engineering effort

Automatically scales deep learning models

Overview

Startup [Gan.AI](#) uses generative AI to create personalized videos with perfect audio, lip-sync, and custom visuals. The company has been using Amazon Web Services (AWS) since its launch in 2021, and recently implemented AWS machine learning fully managed services for real-time video generation.

Using [Amazon SageMaker](#) with multi-model endpoints, and [AWS Lambda](#), Gan.AI can now create thousands of videos simultaneously with ultra-low latency. The business has unlocked new use cases internally and has increased the stickiness of existing customers.



Opportunity | Attracting New Customers with Low-Latency Video

Gan.AI launched in 2021 with a bold mission: to democratize personal video production with generative artificial intelligence (generative AI). The founders wanted to empower everyone, from singular real-estate agents to global enterprises, to easily create hyper-personalized messages for targeted, meaningful marketing. Today, even sports teams such as the San Antonio Spurs and Mumbai Indians use Studio by Gan.AI—the company's main product—to drive fan engagement.

In its initial iteration, Studio was utilized by large consumer brands. Companies would provide a list of recipients to target, and Gan.AI would use this data to generate a custom video for each one, including perfect audio, lip-sync, and visuals. This process could take up to 4 minutes per video.



Then, a new self-serve use case emerged for real-time videos at scale. Anupreet Singh, chief revenue officer at Gan.AI, explains, “Some of our largest customers realized if their users could create their own videos, they could attract people who were otherwise not as engaged with the brand. It was a good opportunity for them to build the top of the marketing funnel and collect data to further engage potential customers.”

Gan.AI faced the challenge of significantly reducing latency in video creation. “Every second matters for consumers visiting a brand’s site to make a video; even 2 minutes was too long,” Singh says. Inferencing for its generative AI models was the primary bottleneck in Gan.AI’s infrastructure. The company sought an efficient, low-latency solution to scale its deep learning models.

Solution | Accelerating Inferencing with Multi-Model Endpoints

Gan.AI had already been using Amazon Web Services (AWS) for several years, including using [Amazon Elastic Container Service](#) (Amazon ECS) with [AWS Lambda](#) to host its machine learning models on CPUs. Following discussions with its AWS account team, the company chose [Amazon SageMaker](#) to inference its deep learning models on GPU instances. In particular, the use of multi-model endpoints would offer the speed required for large-scale projects in real time.

Suvrat Bhooshan, chief executive officer at Gan.AI, says, “Amazon SageMaker helped us host multi-model endpoints without a high cold-start latency of 5–6 minutes. Now, we can serve multiple content requests within 30–40 seconds and scale without much engineering effort.”

Gan.AI spent nearly three months in testing, and with each challenge it encountered along the way, AWS was there to help. For example, when the team needed to ensure quick auto scaling of multi-model endpoints, AWS provided guidance on architecting in such a way that a single GPU instance could be used to serve up to five requests concurrently, responding within a few seconds. Gan.AI typically deploys 8–10 instances that are always running, and that number can scale to about 200 instances when required.

“New users of Amazon SageMaker should thoroughly test and experiment with its various features and GPU instance types before implementation,” Bhooshan advises. “For example, we found that moving inference processes to AWS Lambda enabled more efficient utilization of Amazon SageMaker resources. It’s really useful for scaling in a business such as ours with a mix of deep learning and CPU-intensive workloads.”



Outcome | Scaling Hundreds of Generative AI Models Concurrently

Gan.AI can now simultaneously inference and scale hundreds of custom generative AI models with ultra-low latency. With up to 3,000 concurrent end users each generating custom videos in less than a minute, the Gan.AI framework can produce a million unique videos in just 3 hours. As a component of its real-time offering, Gan.AI also offers brands a live chatbot, complete with custom audio.

Companies across the globe are using Gan.AI Studio to create personalized and innovative customer engagement campaigns. Agoda, for example, partnered with Gan.AI to develop a [personalized pre-roll YouTube ad campaign](#) featuring Indian actor and singer Ayushmann Khurrana. Using Studio, Agoda crafted more than 200 20-second ads where Ayushmann directly interacted with viewers using messages tailored to their dream vacation destinations. The campaign resulted in over 100 million impressions, 10 million views, and 2.5 times increase in brand awareness.

Sahil Sharma, brand lead for India market at Agoda, says, “Agoda’s AI campaign took personalization at scale to the next level by using Gan.AI’s speech generation, lip syncing, and image personalization technology. Gan.AI’s technology not only enables high-quality, rapid results, but also sets a new benchmark for generative AI usage across various categories, including online travel agencies.”

For Gan.AI, real-time video capabilities have unlocked new revenue opportunities for brands, which increases the stickiness of its customer base. The company is currently evaluating additional industry verticals to approach with its solution and continues to work with AWS on new projects to grow the business. These projects include innovative developments like Avatar Zero Shot Models, which incorporate [text-to-speech](#) and voice cloning technology to replicate human speech patterns with synchronized lip movements, and a conversational AI platform designed to offer natural and engaging user interactions.

Learn More

To learn more, visit aws.amazon.com/ai/generative-ai.

About Gan.AI

Gan.AI is a software provider that uses generative AI to create real-time, personalized videos with perfect voice and lip-sync. Large and small businesses worldwide use Gan.AI to generate millions of hyper-customized videos that drive brand engagement and increase conversion rates.

AWS Services Used

Amazon SageMaker

Amazon SageMaker is a fully managed service that brings together a broad set of tools to improve machine learning performance, low-cost machine learning (ML) for any use case.



[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

Amazon Elastic Container Service

Amazon Elastic Container Service (ECS) is a fully managed container orchestration service that helps you to more efficiently deploy, manage, and scale containerized applications.

[Learn more »](#)

More Professional Services Customer Stories

AUSTRALIA



Lawpath Reduces Customer Service Inquiries by 25%...

Lawpath used Anthropic Claude on Amazon Bedrock to develop Lawpath AI, a generative AI solution that increases efficiency and productivity through self-service legal capabilities.

MALAYSIA



123RF Generates AI-Powered Creative Assets...

This case study explores how 123RF, a leading stock media platform, leveraged Amazon Web Services (AWS) to develop a generative AI solution that enables users to create unique images and designs.



2024

2023

ASIA PACIFIC



FarmByte Plants the Seeds of AI-Driven Agriculture...

Learn how FarmByte, a digital agrofood innovator in Malaysia, is revolutionizing agriculture on AWS. By leveraging the FarmByte app, farmers can streamline scheduling and deliveries, significantly improving efficiency. With AI and machine learning integration, FarmByte



2024

AMERICAS



Slalom Gives Customer Agents Data to Solve...

Global technology business Slalom uses generative artificial intelligence (AI) services like Amazon Bedrock to create solutions that improve customer experiences. An Amazon Web Services (AWS) Partner, its focus on the end user helps it deliver truly useful AI use cases.

2024

1 2 3 4 5 6 ... 28

Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Resources for AWS

Developers on AWS

Getting Started

Developer Center



Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |
- Español |
- Français |
- Italiano |
- Português |
- Tiếng Việt |
- Türkçe |
- Русский |
- ไทย |
- 日本語 |
- 한국어 |
- 中文 (简体) |
- 中文 (繁體)

